Boosted Regression Trees (BRTs): Building on what you get wrong

Grant Vagle, Jeremiah Shrovnal, Lynn Waterhouse

Overview of this section

- Boosted regression trees (aka gradient boosted models)
  - Overview of the boosting algorithm
  - Tuning parameters for BRTs
- Deeper model interpretation with SHAP values
- Hands-on coding example(s)

When to use which model? Good question!

Need to see each individual split? CART

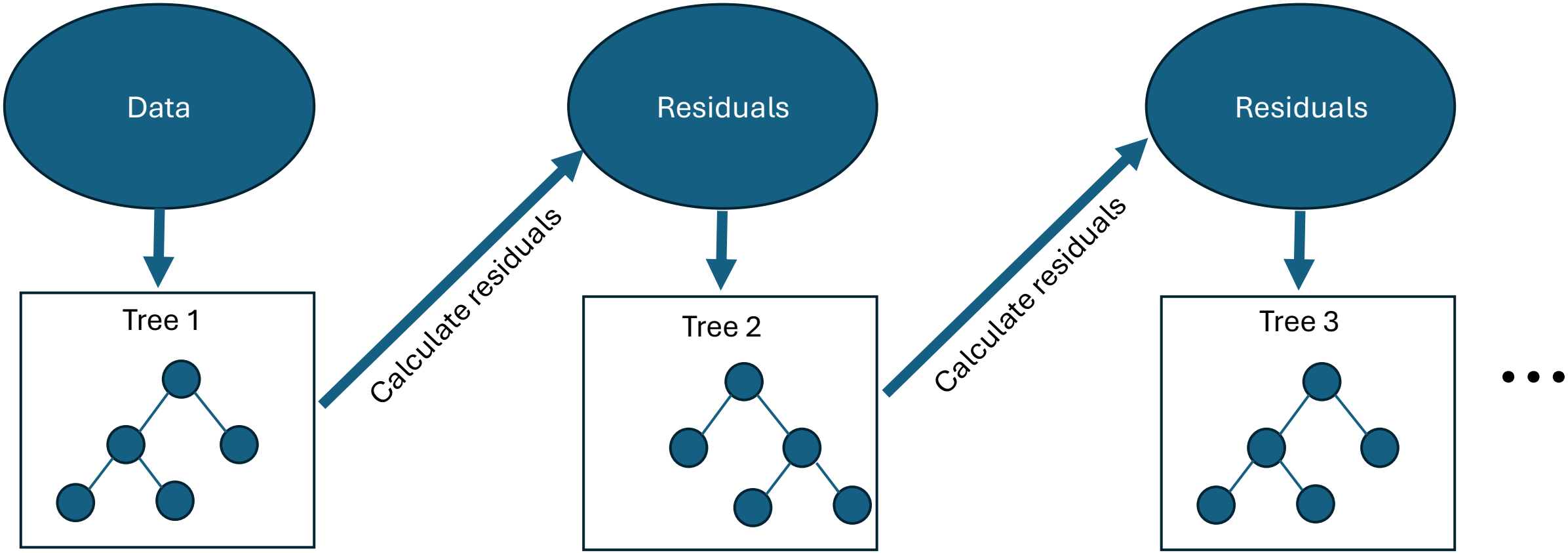
Otherwise, you'll get better predictive performance from:

- Random forests (but can't handle NAs)
- Boosted models (we'll talk about next)
- Neural networks and kin (won't talk about)

Data hungry

Boosting, explained

Each subsequent tree is trained on the residuals from the previous tree



Types of boosted models

- Boosted regression trees (aka gradient boosted models)
- AdaBoost
- XGBoost
- LightGBM
- CatBoost

Differences:

- Regularization (to avoid over-fitting)
- Speed with certain data types

Overview resource:
https://www.geeksforgeeks.org/machine-learning/gradientboosting-vs-adaboost-vs-xgboost-vs-catboost-vs-lightgbm/

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Elith et al. 2008. A working guide to boosted regression trees. Journal of Animal Ecology 77(4): 802-813. https://doi.org/10.1111/j.1365-2656.2008.01390.x

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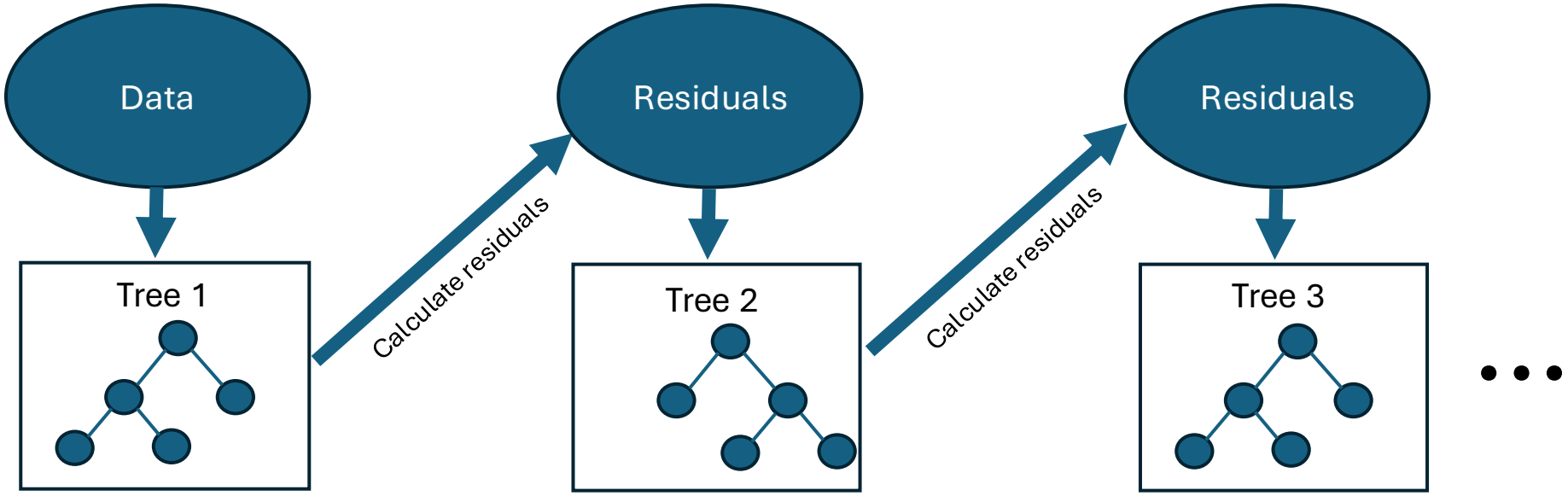
Boosted regression trees

Hyper-parameters:

- Learning rate
- Tree complexity
- Bag fraction
- Number of trees



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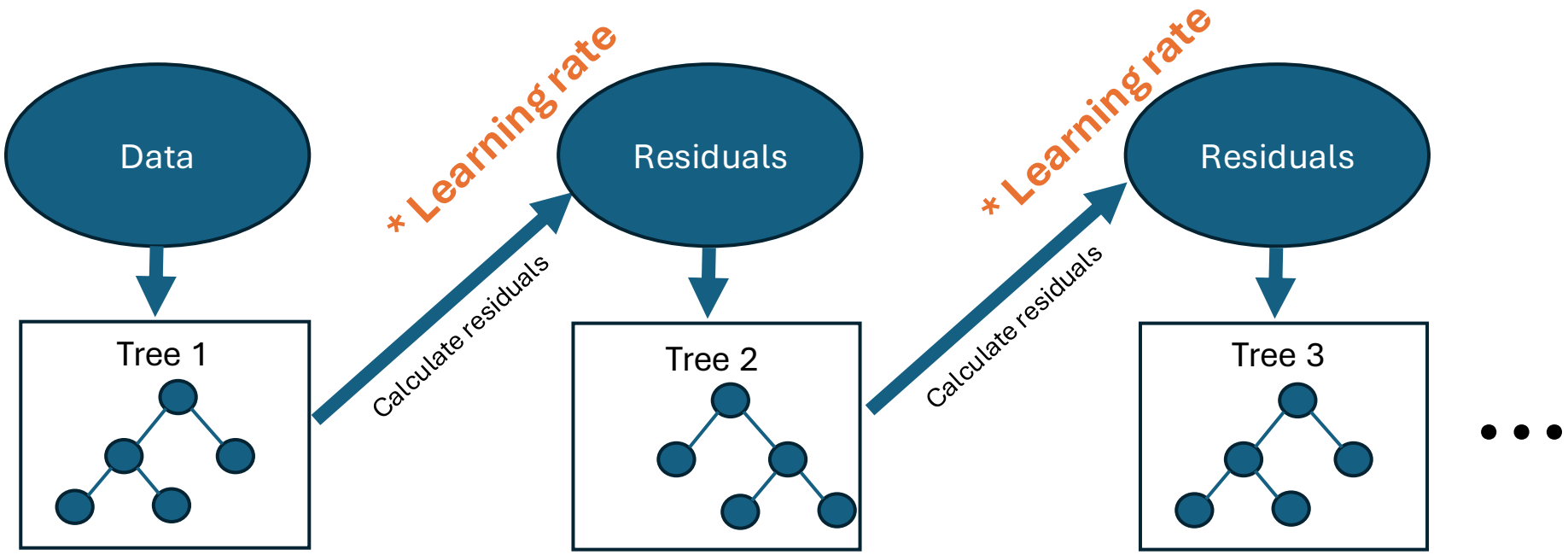

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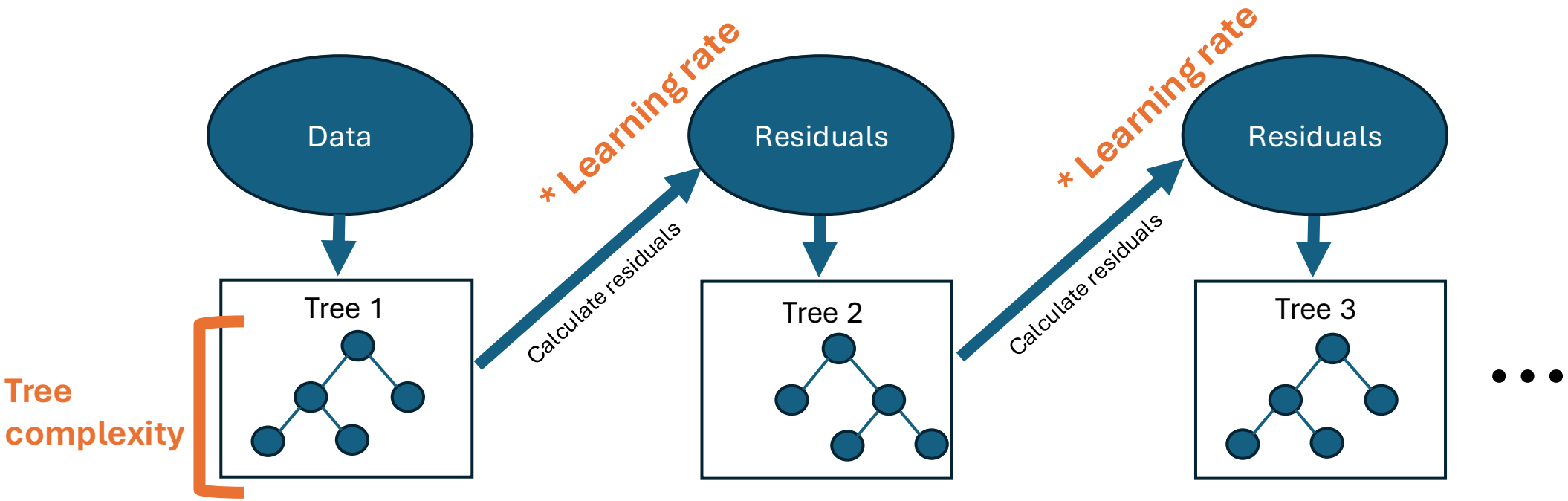

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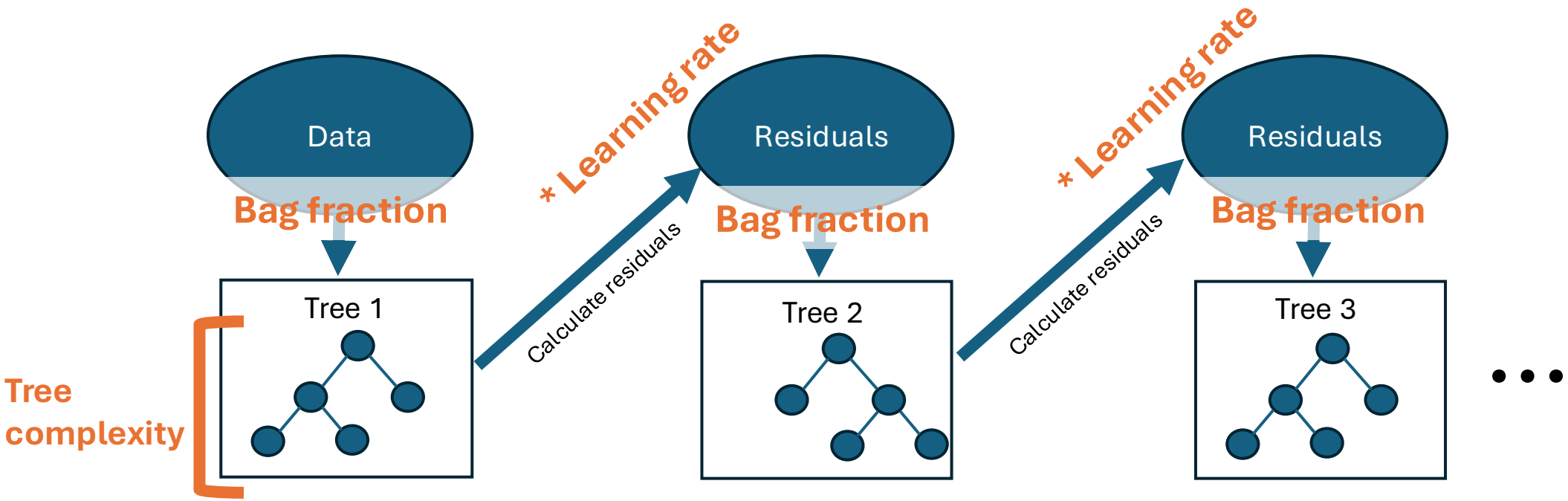

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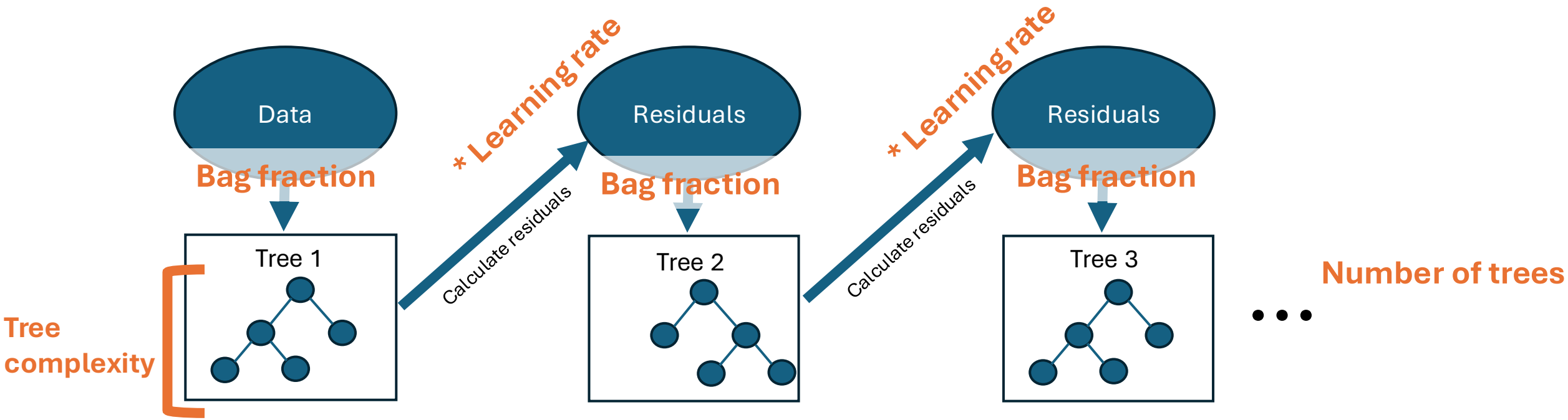
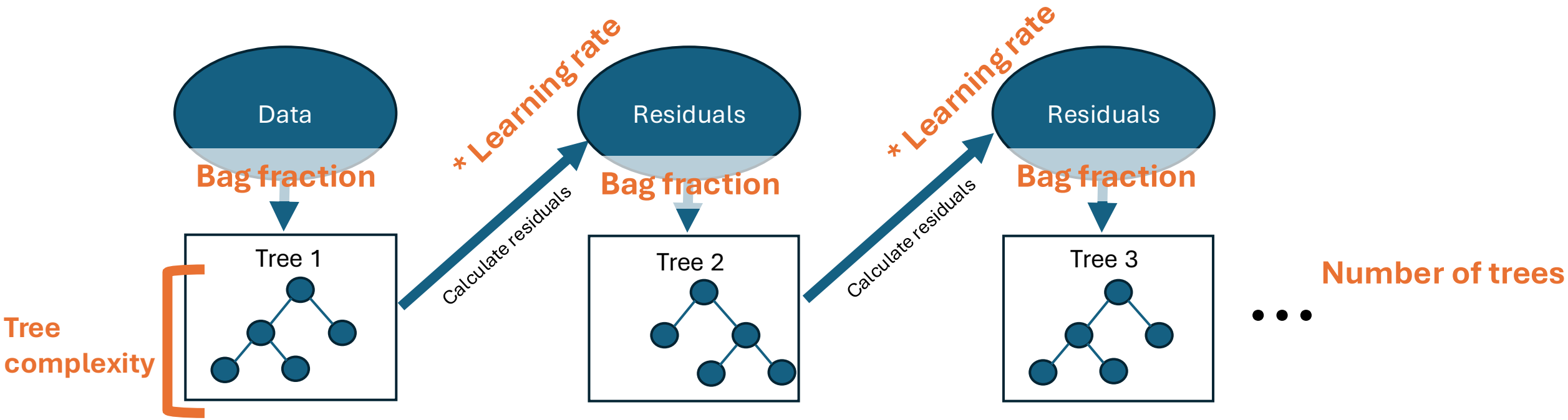


Boosted regression trees

Reducing overfitting

- Learning rate: limits influence of individual tree
- Tree complexity: limits ability for one tree to fit data
- Number of trees: don't build too many

- Bag fraction: don't use all the data for each tree



Model interpretation with SHAP values

SHapley Additive exPlanation (SHAP) values

Which variables drive individual predictions in the model?

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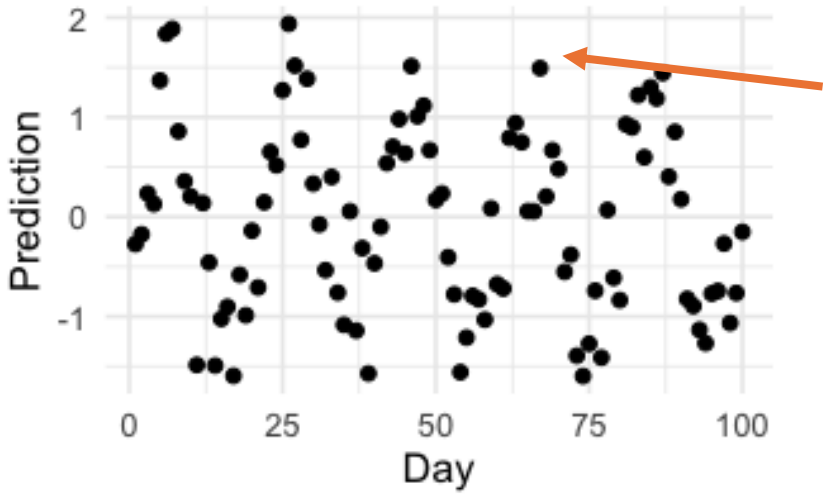

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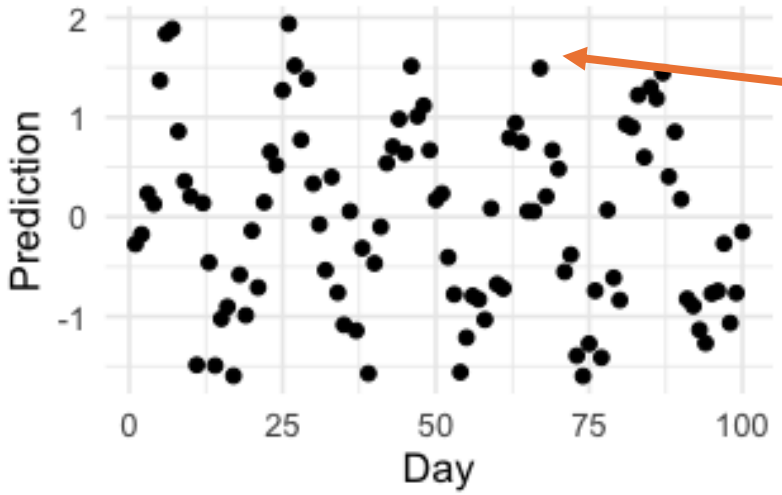

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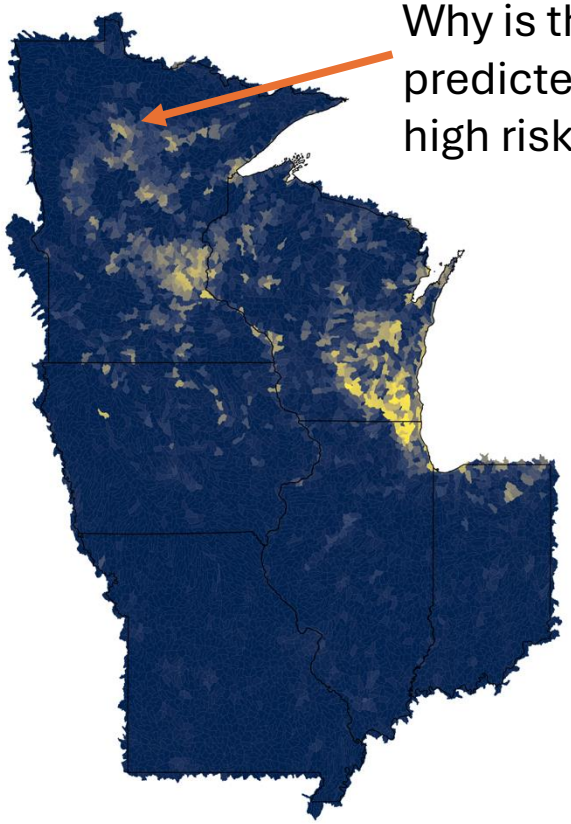

Why is this location predicted to be high risk?

Free ebook on other model interpretation methods: https://christophm.github.io/interpretable-ml-book/

The game theory behind Shapley values

Who’s going to pay for that taxi?

Molnar, C. 2023. Interpreting machine learning with SHAP.

The game theory behind Shapley values

Who's going to pay for that taxi?

- First, we consider the cost of the taxi for all combinations of riders

Passengers	Cost	
{}	$0	
Alice	$15	
Bob	$25	Luxury taxis
Charlie	$38	Lives further away
Alice, Bob	$25	Alice & Bob live together
Alice, Charlie	$41	
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The game theory behind Shapley values

Who's going to pay for that taxi?

- First, we consider the cost of the taxi for all combinations of riders
- Next, we consider the marginal contribution for adding a person to each of these coalitions
- Then, we calculate the **weighted average of the marginal contributions** to all possible coalitions

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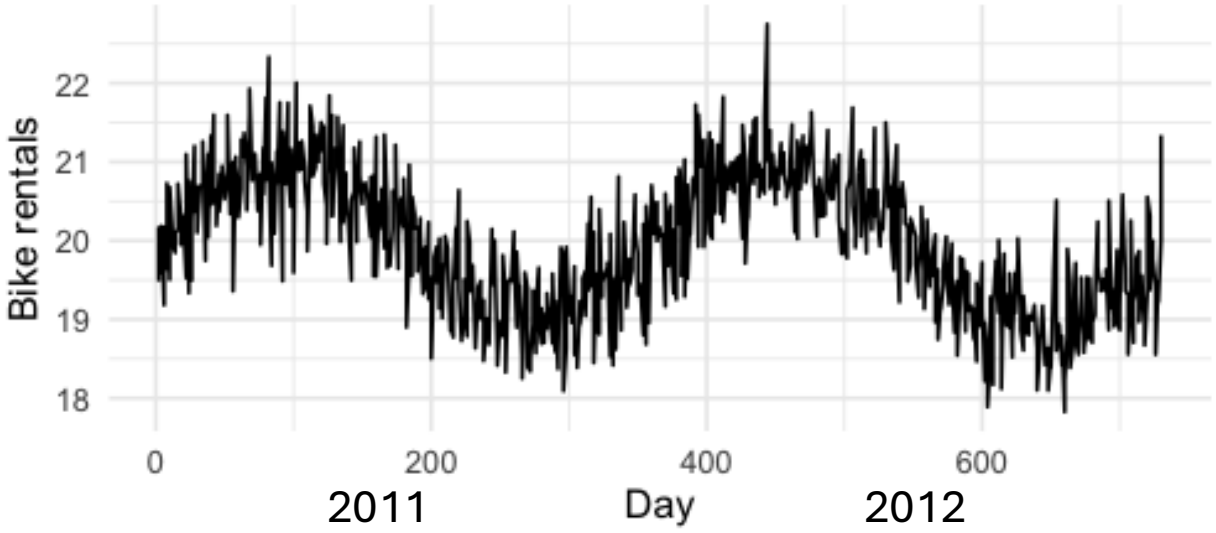
Our example: predicting bike rentals

Daily bike rentals ~ season + weather + day of week



*Not the real data

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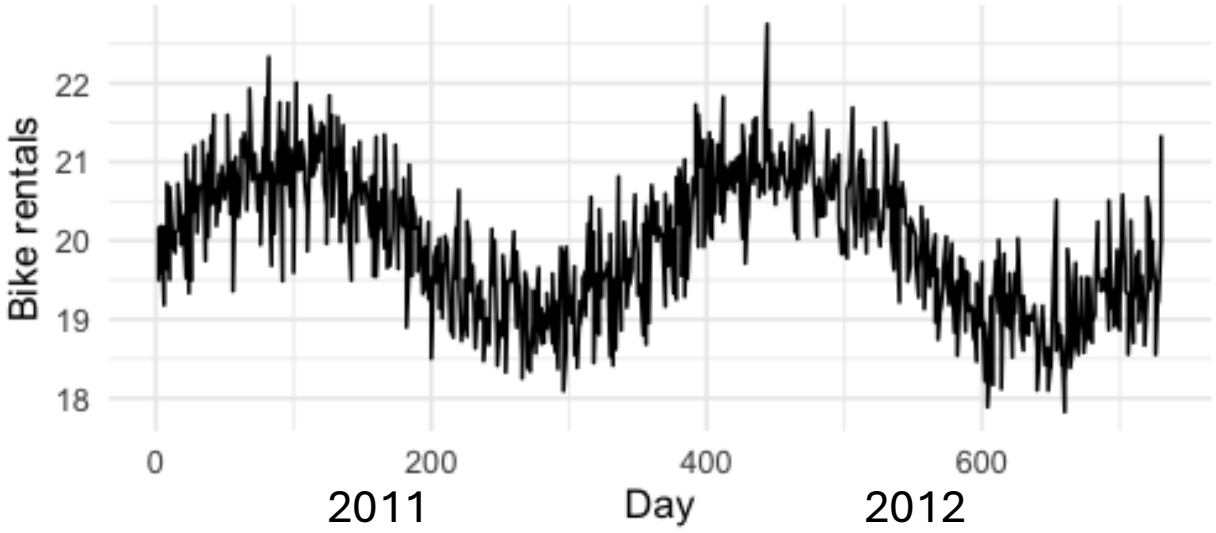
Prediction for day 446 (March 22, 2012)



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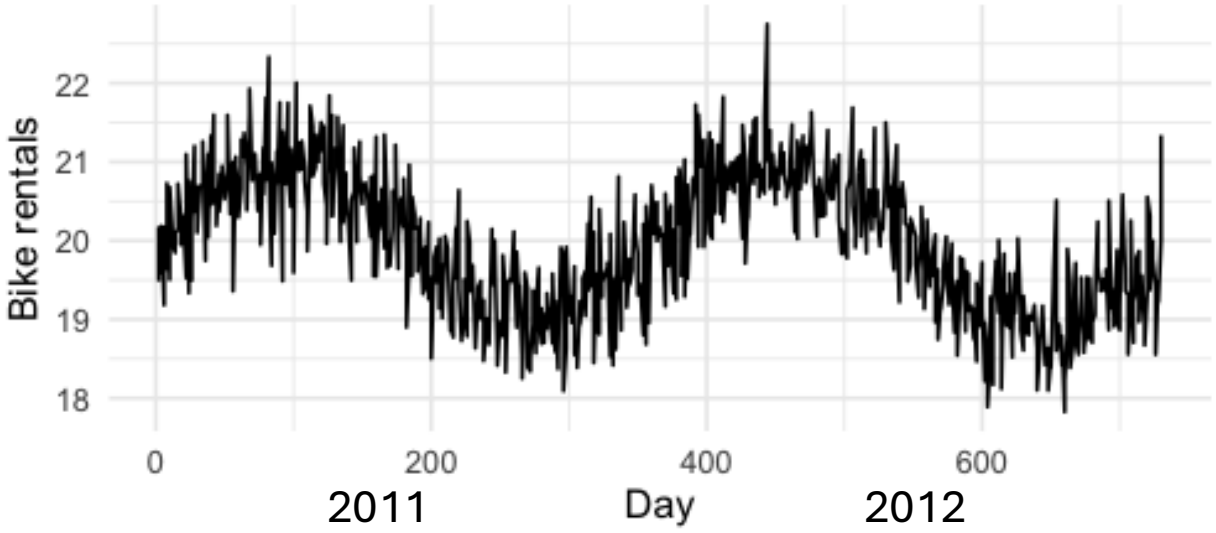

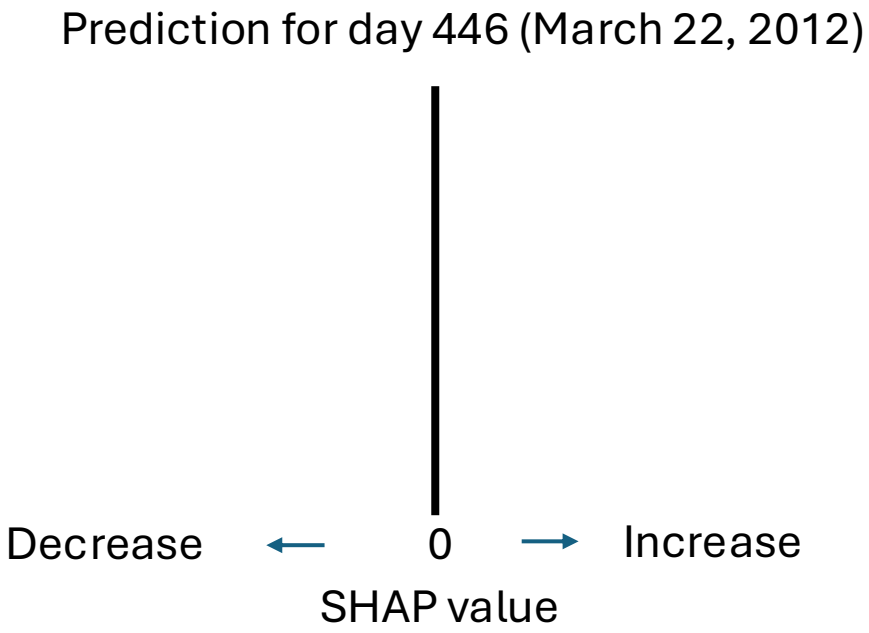

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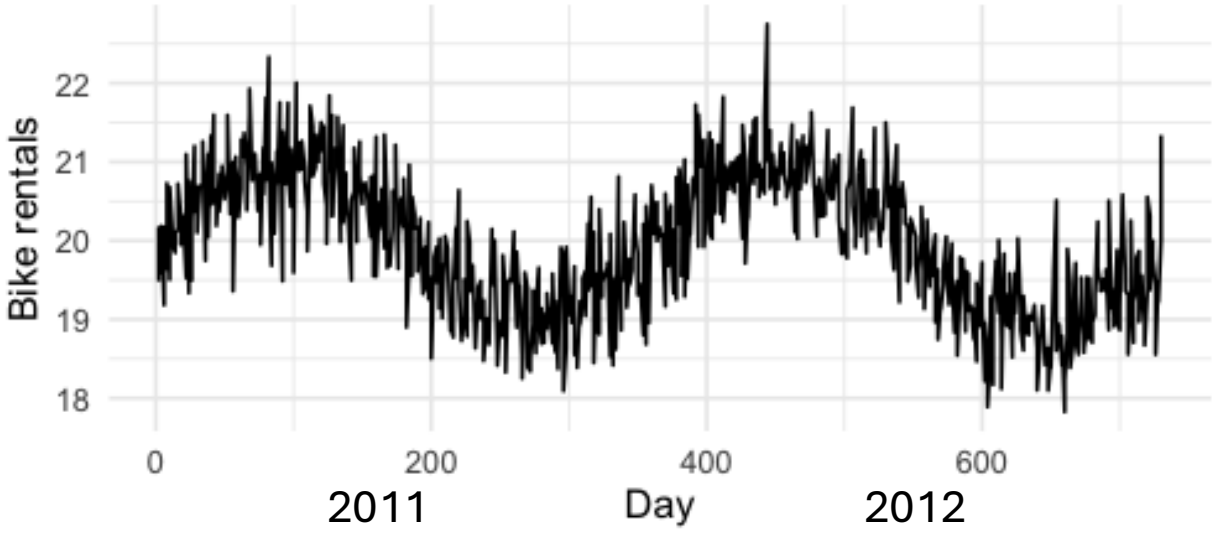

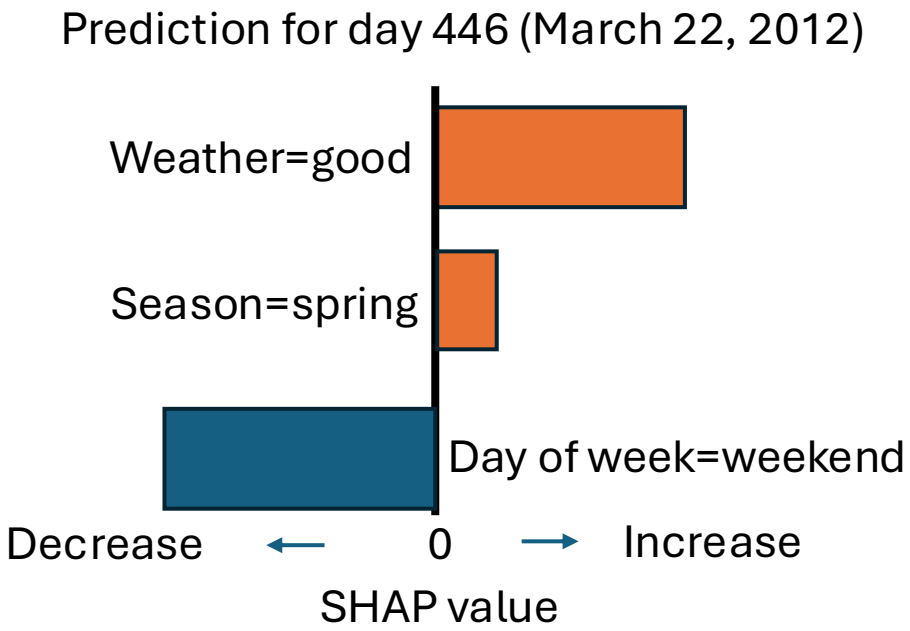

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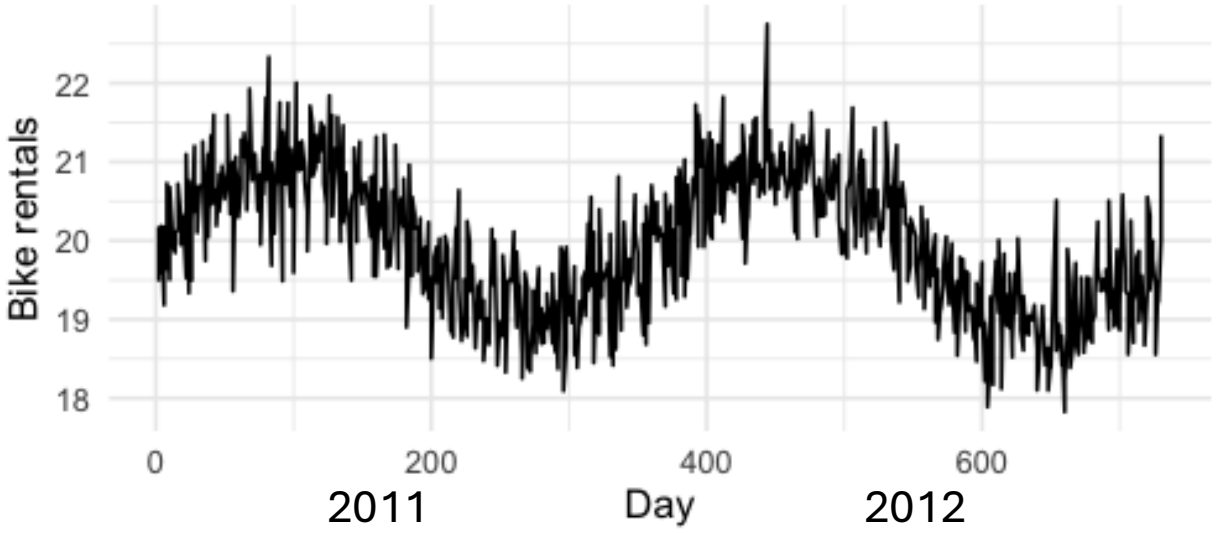

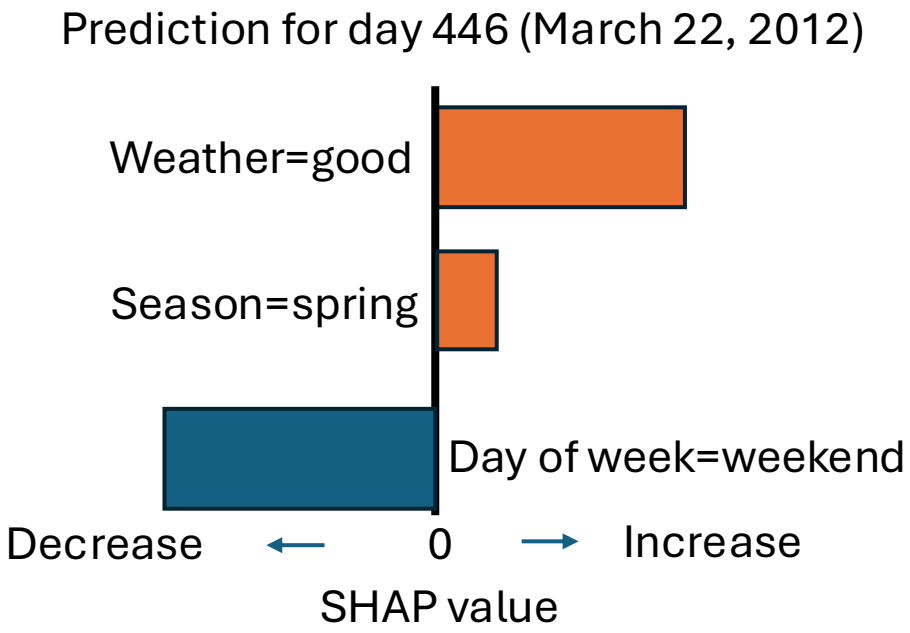

Many possible summaries of SHAP values across the dataset!

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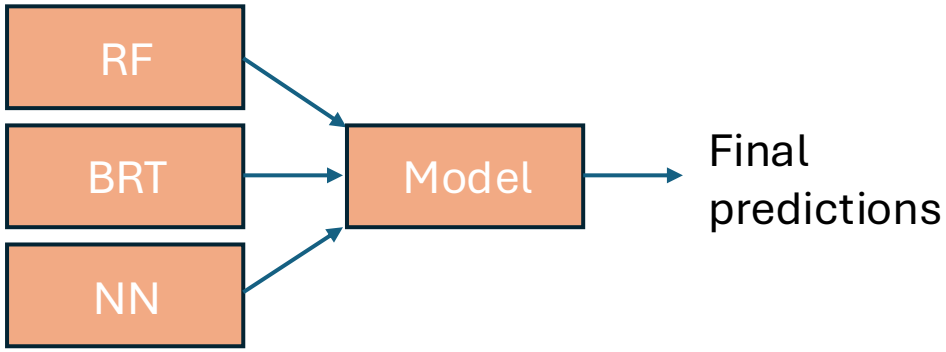
Also, ensemble models work well too!